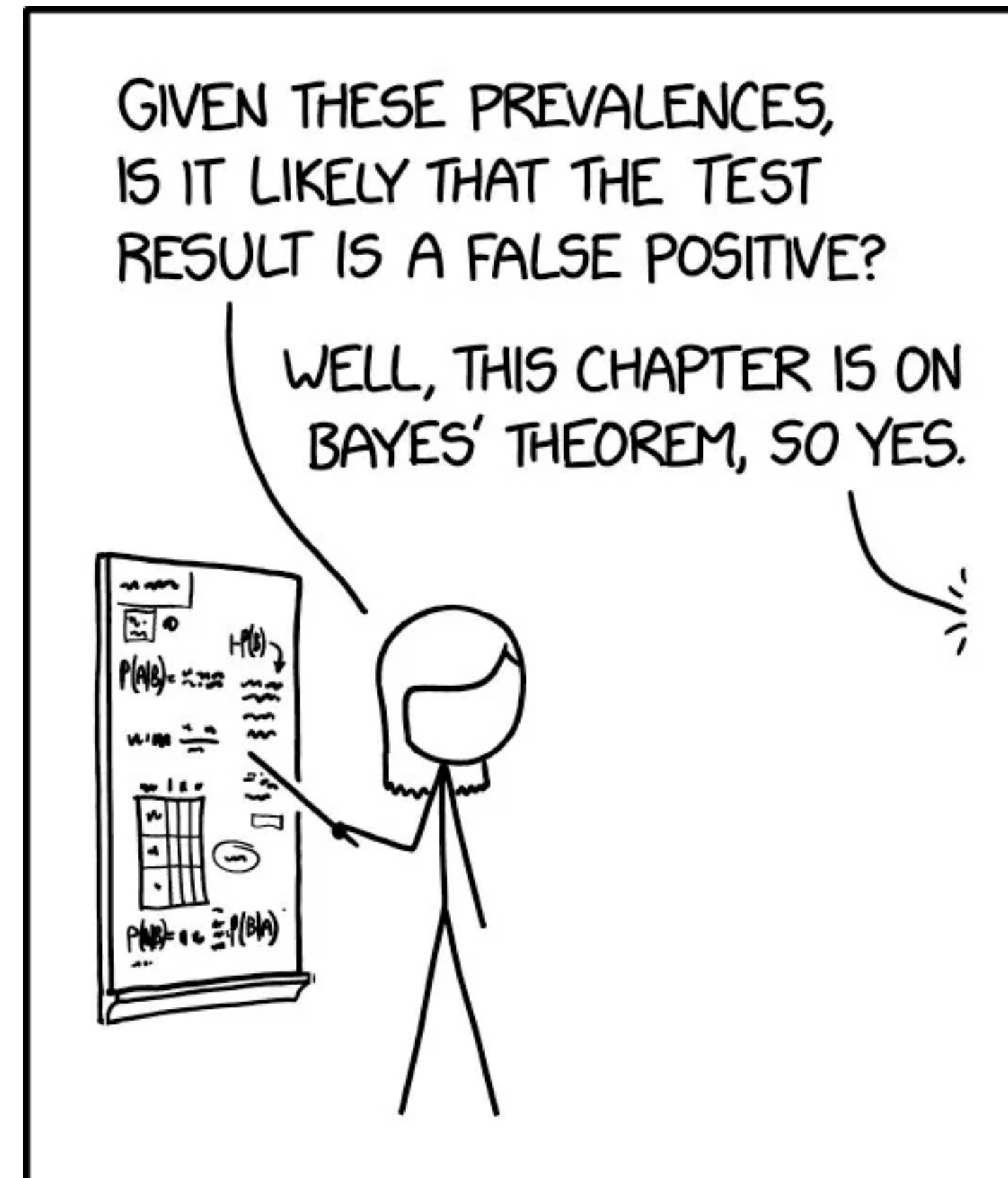
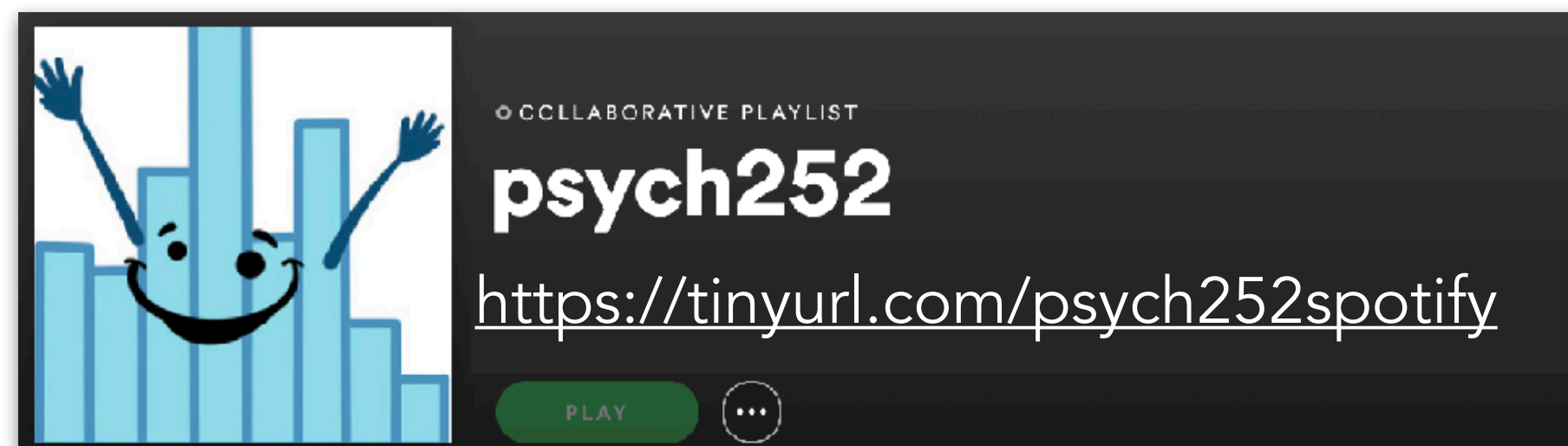


Bayesian data analysis 5



SOMETIMES, IF YOU UNDERSTAND
BAYES' THEOREM WELL ENOUGH,
YOU DON'T NEED IT.



03/09/2026

Logistics

Final presentation survey → goal: everyone!

Questions Responses **21** Settings

Final presentation

Thanks for filling out this survey to help us with planning!

How are you planning to present? *

☐ In class (preferred option if possible)

☐ Remotely (live)

☐ I will record the presentation and submit a video before March 15th.

☐ Other...

What's your name (e.g. Tobias Gerstenberg)? *

Short answer text

What's the name of your team's github repository (e.g. final-project-tobi)? *

Short answer text

How many people are in your team (e.g. 1, 2, or 3)?

Short answer text

<https://tinyurl.com/psych252presentation26>

Plan for today

- Quick recap
- Going beyond
 - I want only positive coefficients
 - Dealing with unequal variance
- A worked example

Quick recap

Quick recap: Sleep data

1. Specify and fit the model

```
1 fit.brm_sleep = brm(formula = reaction ~ 1 + days + (1 + days | subject),  
2 data = df.sleep,  
3 seed = 1,  
4 file = "cache/brm_sleep")
```

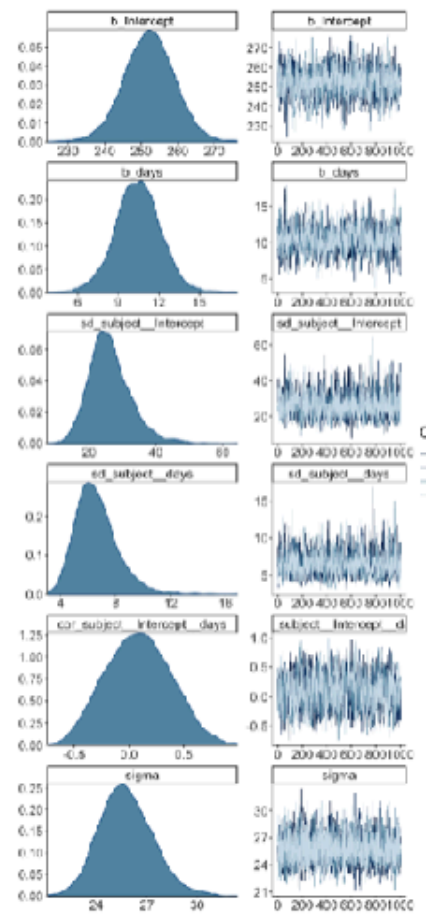


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a) Did the inference work?

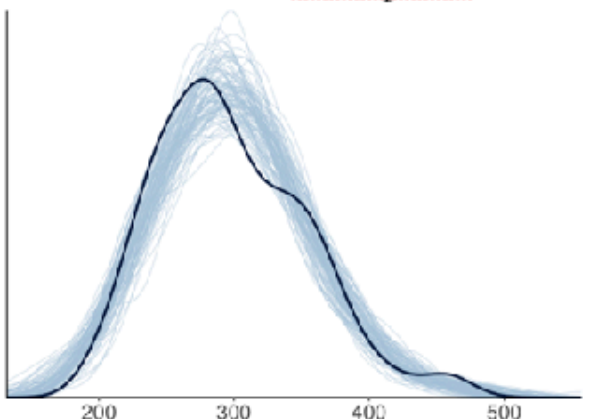
```
1 fit.brm_sleep %>%  
2 plot(N = 6)
```

these look good!



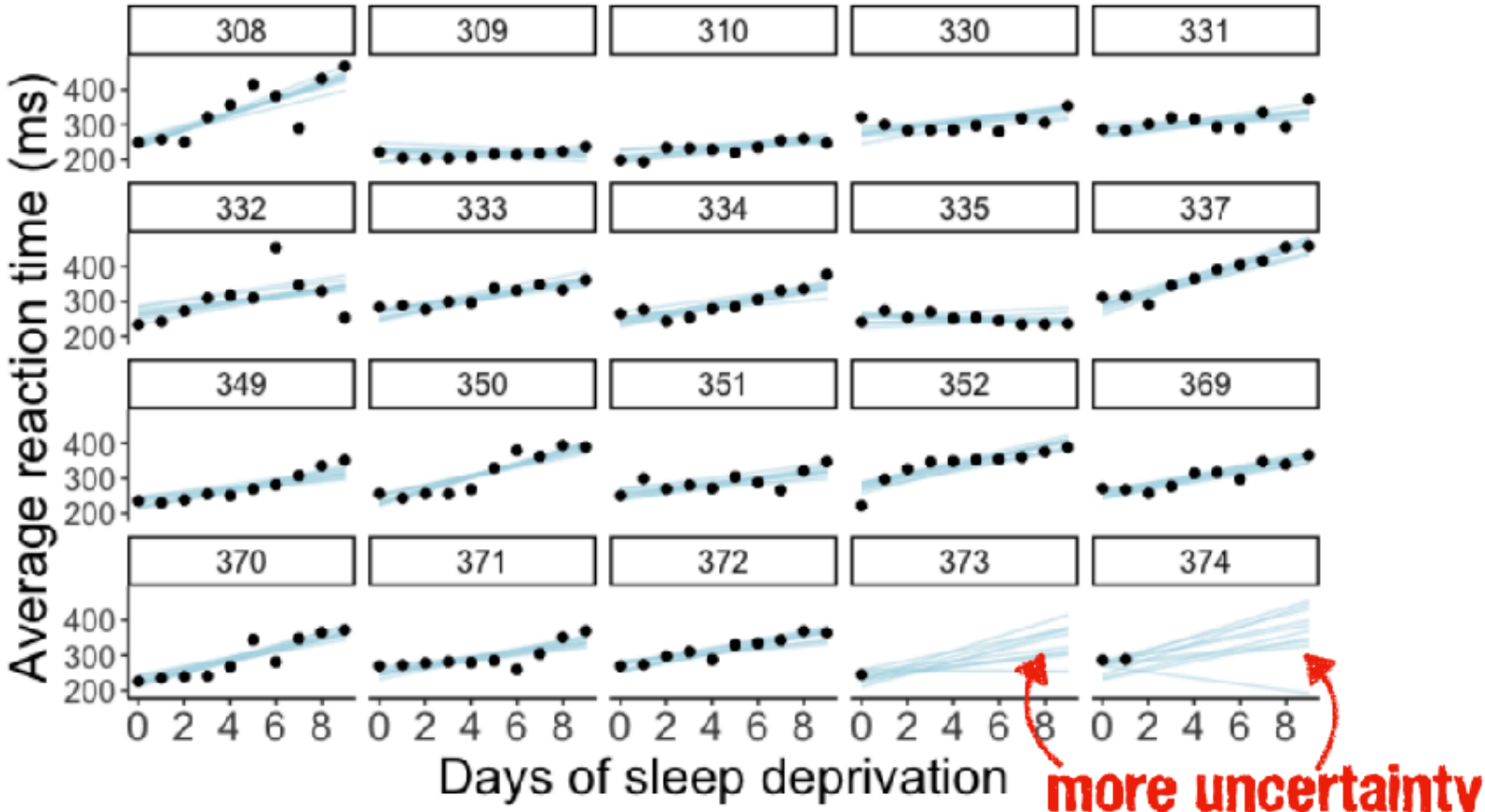
also looks good!

```
1 pp_check(fit.brm_sleep,  
2 nsamples = 100)
```



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b) Visualize the model predictions



10 random samples from the posterior distribution

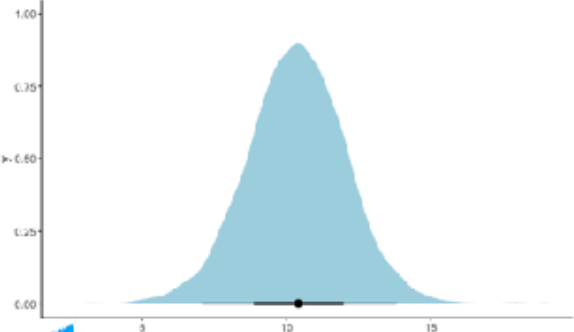
48

5. Test specific hypotheses

Did reaction times increase with the number of days of sleep deprivation?

```
1 fit.brm_sleep %>%  
2 summary()
```

Family: gaussian									
Links: mu = identity; sigma = identity									
Formula: reaction ~ 1 + days + (1 + days subject)									
Data: df.sleep (Number of observations: 183)									
Samples: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;									
total post-warmup samples = 4000									
Group-Level Effects:									
~subject (Number of levels: 20)									
	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk	ESS	Tail	ESS
sd Intercept	25.16	6.25	15.65	46.54	1.00	1879	2463		
sd days	6.25	1.52	4.14	10.13	1.00	1145	1625		
cor Intercept, days	0.35	0.25	0.46	0.67	1.00	903	1526		
Population-Level Effects:									
	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk	ESS	Tail	ESS
Intercept	252.18	6.46	239.47	265.42	1.00	1826	2766		
days	10.46	1.65	7.13	13.78	1.00	1203	1782		
Family Specific Parameters:									
	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk	ESS	Tail	ESS
sigma	25.77	1.57	22.63	29.14	1.00	3864	2773		



54

6

Quick recap: Titanic data

1. Specify and fit the model

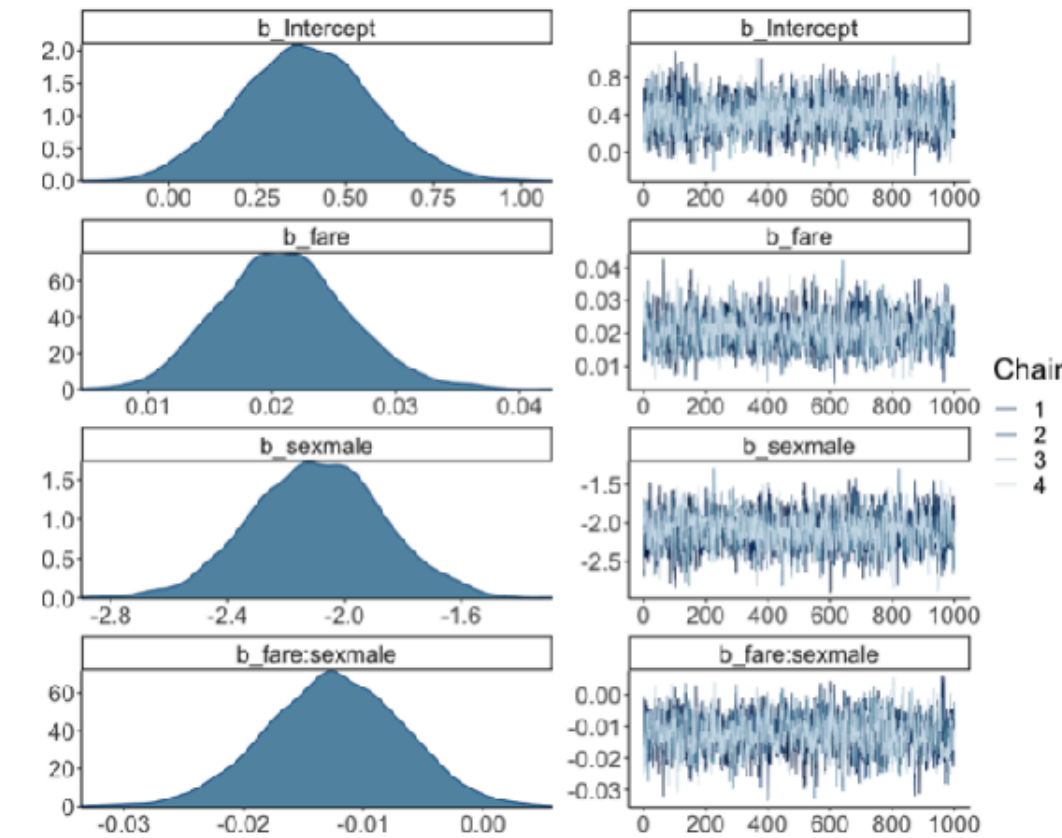
```
1 fit.brm_titanic = brm(formula = survived ~ 1 + fare * sex,  
2                       family = "bernoulli",  
3                       data = df.titanic,  
4                       file = "cache/brm_titanic",  
5                       seed = 1)
```

just need to
change the family

63

a) Did the inference work?

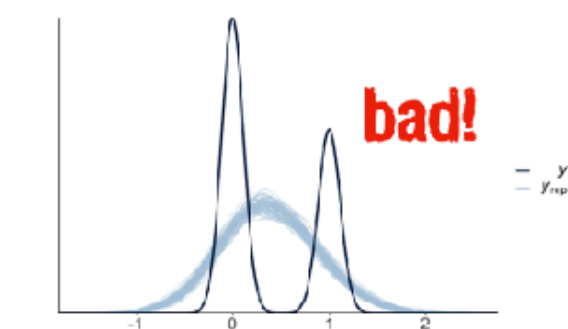
```
1 fit.brm_titanic %>%  
2 plot()
```



```
1 pp_check(fit.brm_titanic,  
2          nsamples = 100)
```



model with Gaussian family



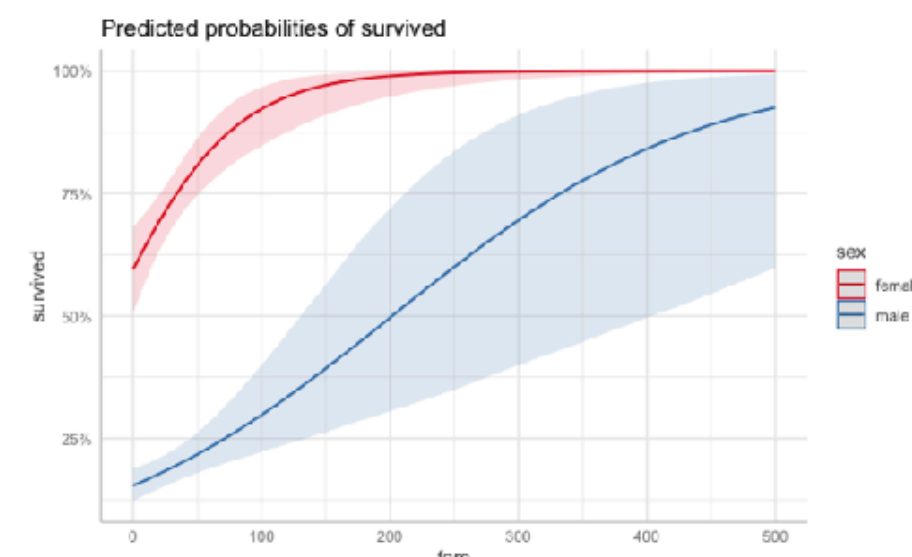
66

4. Interpret the model parameters



```
1 fit.brm_titanic %>%  
2 ggpredict(terms = c("fare [0:500]", "sex"))
```

```
# Predicted probabilities of survived  
# x = fare  
  
# sex = female  
x | Predicted | 95% CI  
0 | 0.60 | [0.51, 0.68]  
83 | 0.89 | [0.82, 0.95]  
167 | 0.98 | [0.93, 1.00]  
250 | 1.00 | [0.97, 1.00]  
333 | 1.00 | [0.99, 1.00]  
500 | 1.00 | [1.00, 1.00]  
  
# sex = male  
x | Predicted | 95% CI  
0 | 0.15 | [0.12, 0.19]  
83 | 0.27 | [0.21, 0.35]  
167 | 0.43 | [0.28, 0.62]  
250 | 0.60 | [0.35, 0.84]  
333 | 0.75 | [0.43, 0.94]  
500 | 0.93 | [0.60, 0.99]
```



70

5. Test specific hypotheses

Was the effect of fare on survival different for men vs women?

```
1 fit.brm_titanic %>%  
2 emtrends(specs = pairwise ~ sex,  
3          var = "fare")
```

```
$emtrends  
sex    fare.trend lower.HPD upper.HPD  
female 0.02083    0.01129    0.0316  
male    0.00845    0.00385    0.0135  
  
Point estimate displayed: median  
HPD interval probability: 0.95  
  
$contrasts  
contrast estimate lower.HPD upper.HPD  
female - male 0.0124 0.000884 0.0232  
  
Point estimate displayed: median  
HPD interval probability: 0.95
```

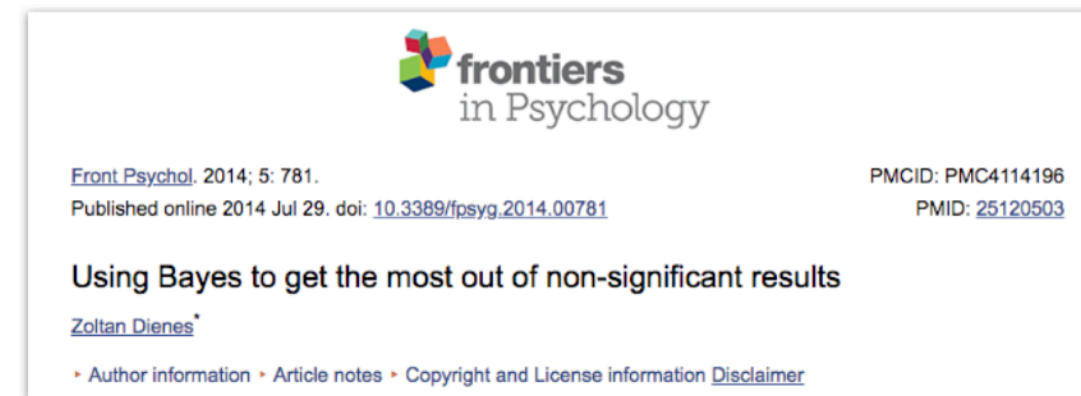
the chance of survival
increased more with fare
for female than male
passengers

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Quick recap: Evidence for the null hypothesis

Evidence for the null hypothesis



[HTML] Using Bayes to get the most out of non-significant results
Z. Dienes - Frontiers in psychology, 2014 - frontiersin.org
No scientific conclusion follows automatically from a statistically non-significant result, yet people routinely use non-significant results to guide conclusions about the status of theories (or the effectiveness of practices). To know whether a non-significant result counts against a theory, or if it just indicates data insensitivity, researchers must use one of: power, intervals (such as confidence or credibility intervals), or else an indicator of the relative evidence for one theory over another, such as a Bayes factor. I argue Bayes factors allow theory to be ...
☆ Cited by 966 Related articles All 14 versions Web of Science: 583 Import into BibTeX

2468 now

- There is nothing special about H_0 compared to H_1 in Bayesian inference
- We can get evidence of H_0 over H_1 (e.g. using the Bayes factor approach)

Bayes factor

$$BF_{01} = \frac{p(D|H_0)}{p(D|H_1)}$$

probability of the data given H_0

probability of the data given H_1

check this out

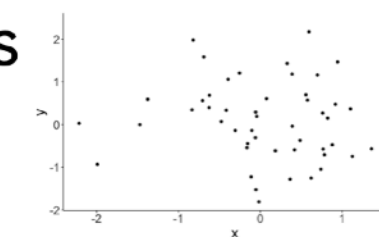
<https://vuorre.com/posts/bayes-factors-with-brms/>

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Evidence for the null hypothesis

```
1 fit.brm_loo1 = brm(formula = y ~ 1, data = df.loo)
2
3 fit.brm_loo2 = brm(formula = y ~ 1 + x, data = df.loo)
4
5 fit.brm_loo1 = add_criterion(fit.brm_loo1, criterion = "loo")
6
7 fit.brm_loo2 = add_criterion(fit.brm_loo2, criterion = "loo")
```



```
loo_compare(fit.brm_loo1, fit.brm_loo2)
```

	elpd_diff	se_diff
fit.brm_loo1	0.0	0.0
fit.brm_loo2	-1.1	0.5

approximate
leave-one-out
cross-validation

Difference in Expected Log Predictive Density
In plain English: It tells you how much worse a model is at predicting new data compared to the best model in your list.

```
model_weights(fit.brm_loo1, fit.brm_loo2)
```

fit.brm_loo1	fit.brm_loo2
99.99999	0.00001

The Rule of Thumb: If the absolute value of `elpd_diff` is more than 2 times larger than the `se_diff`, you can be confident that the winning model is genuinely better.

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8

Going beyond

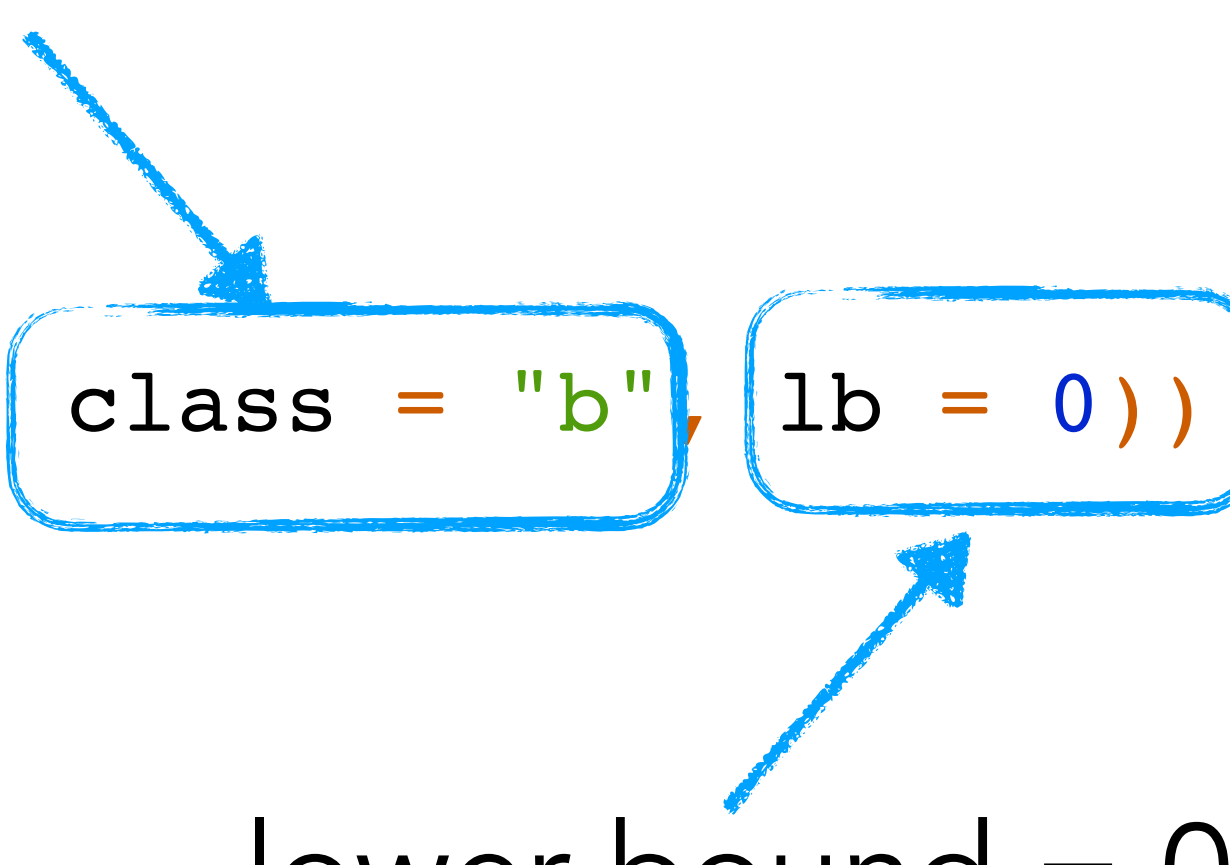
I want only positive coefficients!

I only want positive coefficients!

```
1 brm(formula = how_much_i_love_stats ~ 1 + nilam + tobi + justin + cynthia +  
      verity + shashanka + alvin,  
2     data = df.stats_love)
```

coefficients in the model

```
1 # priors  
2 priors = c(set_prior("normal(0,10)",  
                      class = "b",  
                      lb = 0))
```



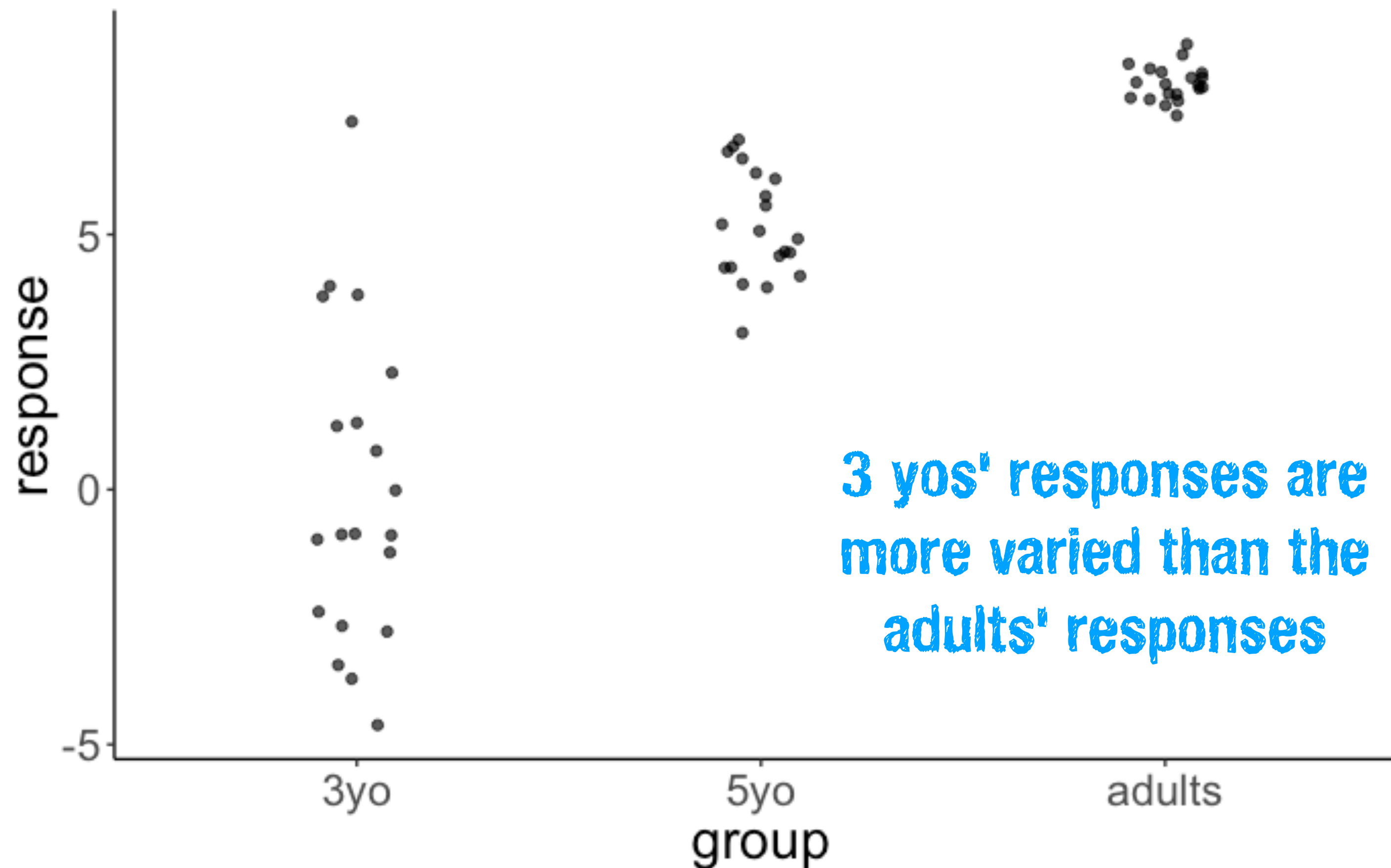
lower bound = 0

```
1 brm(formula = how_much_i_love_stats ~ 1 + nilam + tobi + justin + cynthia +  
      verity + shashanka + alvin,  
2     prior = priors,  
3     data = df.stats_love)
```


Dealing with unequal variance

Unequal variance aka heteroscedasticity

```
1 df.variance = tibble(group = rep(c("3yo", "5yo", "adults"), each = 20),  
2                       response = rnorm(n = 60,  
3                                       mean = rep(c(0, 5, 8), each = 20),  
4                                       sd = rep(c(3, 1.5, 0.3), each = 20)))
```



Unequal variance aka heteroscedasticity

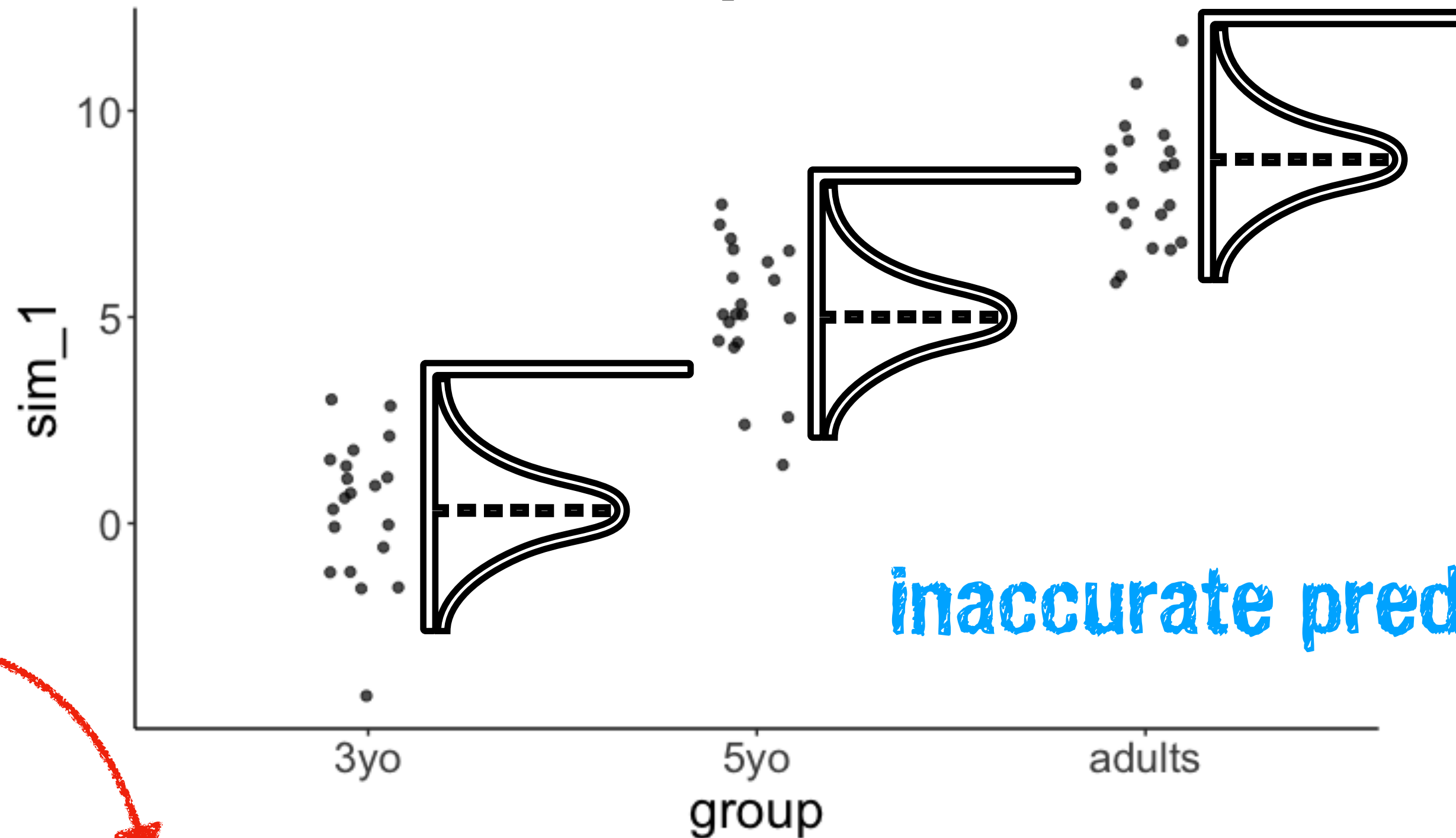
```
1 fit.lm1 = lm(formula = response ~ 1 + group,  
2             data = df.variance)  
3  
4 fit.lm1 %>%  
5   summary()
```

```
Call:  
lm(formula = response ~ 1 + group, data = df.variance)  
  
Residuals:  
      Min       1Q   Median       3Q      Max   
-4.6145 -0.8288 -0.0879  0.6315  7.2193  
  
Coefficients:  
              Estimate Std. Error t value Pr(>|t|)      
(Intercept) -0.005336   0.421618  -0.013    0.99      
group5yo     5.172810   0.596258   8.675 5.25e-12 ***  
groupadults  7.970655   0.596258  13.368 < 2e-16 ***  
---  
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
  
Residual standard error: 1.886 on 57 degrees of freedom  
Multiple R-squared:  0.7635,    Adjusted R-squared:  0.7552  
F-statistic: 91.99 on 2 and 57 DF,  p-value: < 2.2e-16
```

Unequal variance aka heteroscedasticity

```
1 fit.lm1 %>%
2   simulate() %>%
3   bind_cols(df.variance) %>%
4   ggplot(aes(x = group, y = sim_1)) +
5   geom_jitter(height = 0,
6               width = 0.1,
7               alpha = 0.7)
```

only one
standard
deviation
fitted



inaccurate predictions ...

r.squared	adj.r.squared	sigma	statistic	p.value	df	logLik	AIC	BIC	deviance	df.residual
0.76	0.76	1.89	91.99	0	3	-121.65	251.3	259.68	202.65	57

Unequal variance aka heteroscedasticity

```
1 fit.brm1 = brm(formula = bf(response ~ group,  
2                      sigma ~ group),  
3               data = df.variance,  
4               file = "cache/brm1",  
5               seed = 1)
```

← modeling both the
means and variances

```
Family: gaussian  
Links: mu = identity; sigma = log  
Formula: response ~ group  
         sigma ~ group  
Data: df.variance (Number of observations: 60)  
Samples: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;  
         total post-warmup samples = 4000
```

Population-Level Effects:

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	-0.01	0.73	-1.41	1.51	1.01	1107	1072
sigma_Intercept	1.15	0.17	0.85	1.51	1.00	1991	1922
group5yo	5.18	0.77	3.60	6.65	1.00	1252	1327
groupadults	7.98	0.74	6.47	9.37	1.01	1110	1079
sigma_group5yo	-1.05	0.24	-1.51	-0.57	1.00	2249	2420
sigma_groupadults	-2.19	0.24	-2.66	-1.74	1.00	2171	2427

Samples were drawn using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

Unequal variance aka heteroscedasticity

```
Family: gaussian
Links: mu = identity; sigma = log ← on a log scale!
Formula: response ~ group
        sigma ~ group
Data: df.variance (Number of observations: 60)
Samples: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
        total post-warmup samples = 4000
```

Population-Level Effects:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	-0.01	0.73	-1.41	1.51	1.01	1107	1072
sigma_Intercept	1.15	0.17	0.85	1.51	1.00	1991	1922
group5yo	5.18	0.77	3.60	6.65	1.00	1252	1327
groupadults	7.98	0.74	6.47	9.37	1.01	1110	1079
sigma_group5yo	-1.05	0.24	-1.51	-0.57	1.00	2249	2420
sigma_groupadults	-2.19	0.24	-2.66	-1.74	1.00	2171	2427

Samples were drawn using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

```
mean = c(0, 5, 8)
sd    = c(3, 1.5, 0.3)
```

pretty
good!



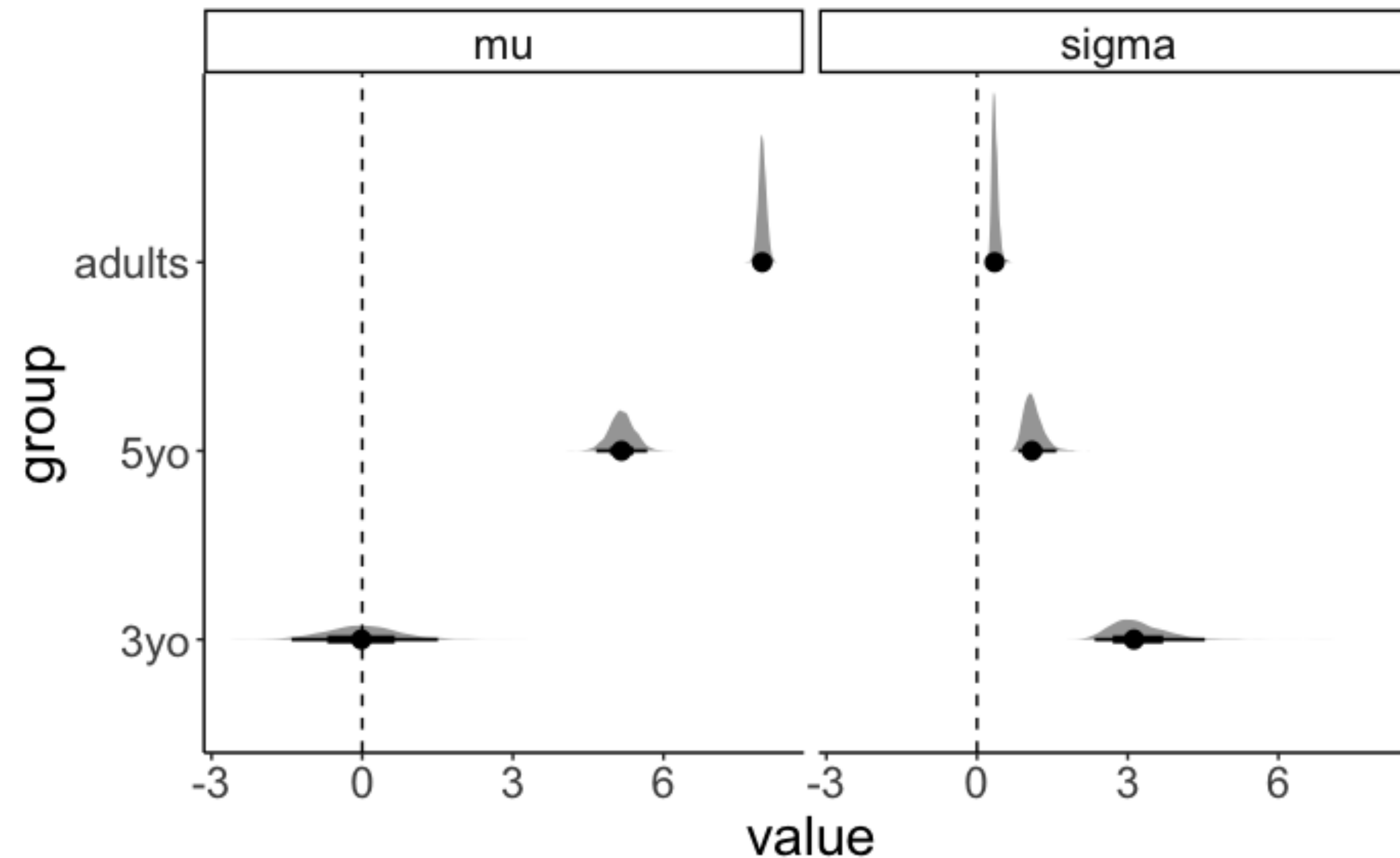
3 year olds $e^{1.15} = 3.16$

5 year olds $e^{1.15+(-1.05)} = 1.10$

adults $e^{1.15+(-2.19)} = 0.35$

Unequal variance aka heteroscedasticity

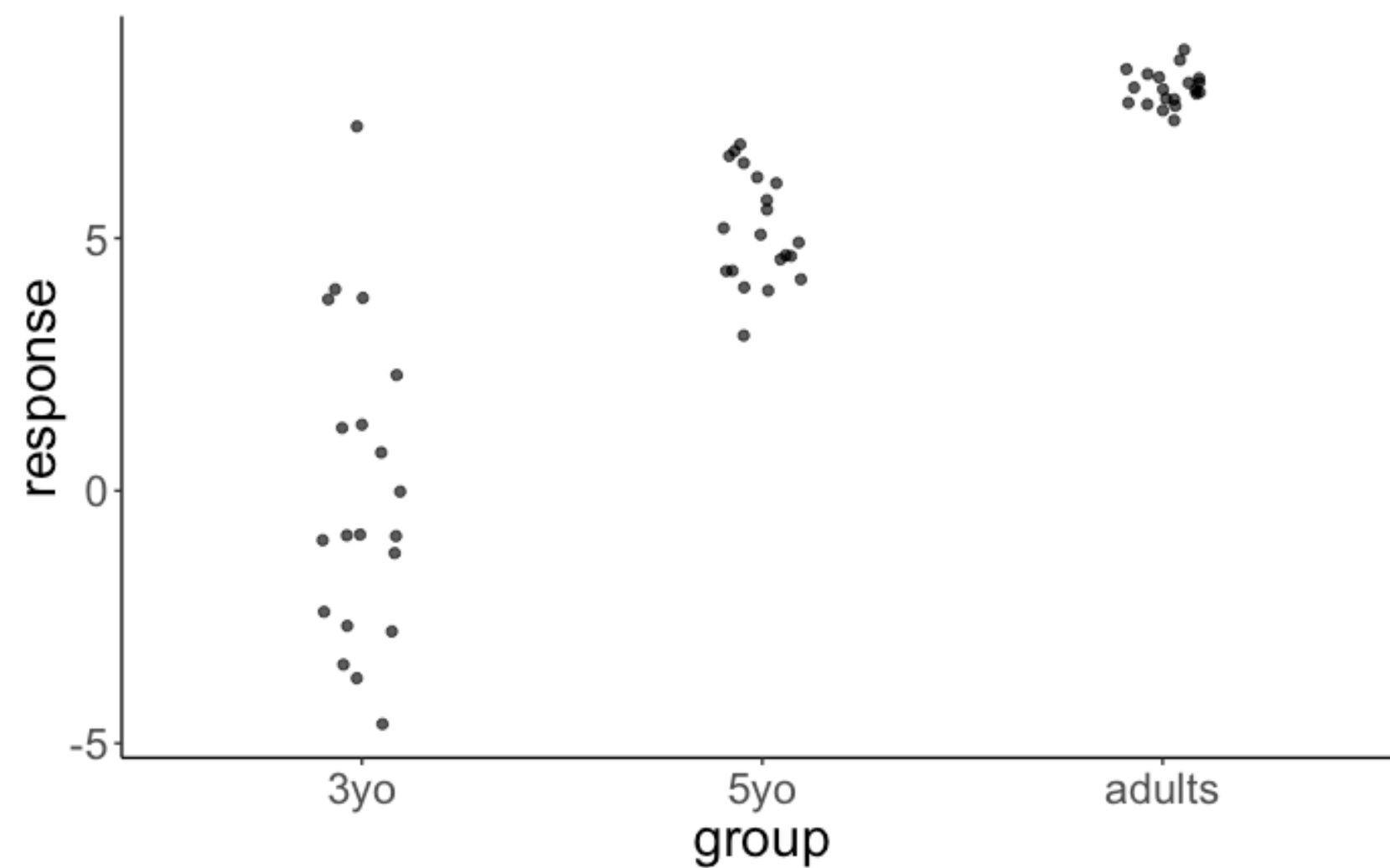
```
1 df.variance %>%
2   expand(group) %>%
3   add_fitted_draws(fit.brm1, dpar = TRUE) %>%
4   select(group, .row, .draw, posterior = .value, mu, sigma) %>%
5   pivot_longer(cols = c(mu, sigma),
6                 names_to = "index",
7                 values_to = "value") %>%
8   ggplot(aes(x = value, y = group)) +
9   geom_halfeyeh() +
10  geom_vline(xintercept = 0, linetype = "dashed") +
11  facet_grid(cols = vars(index))
```



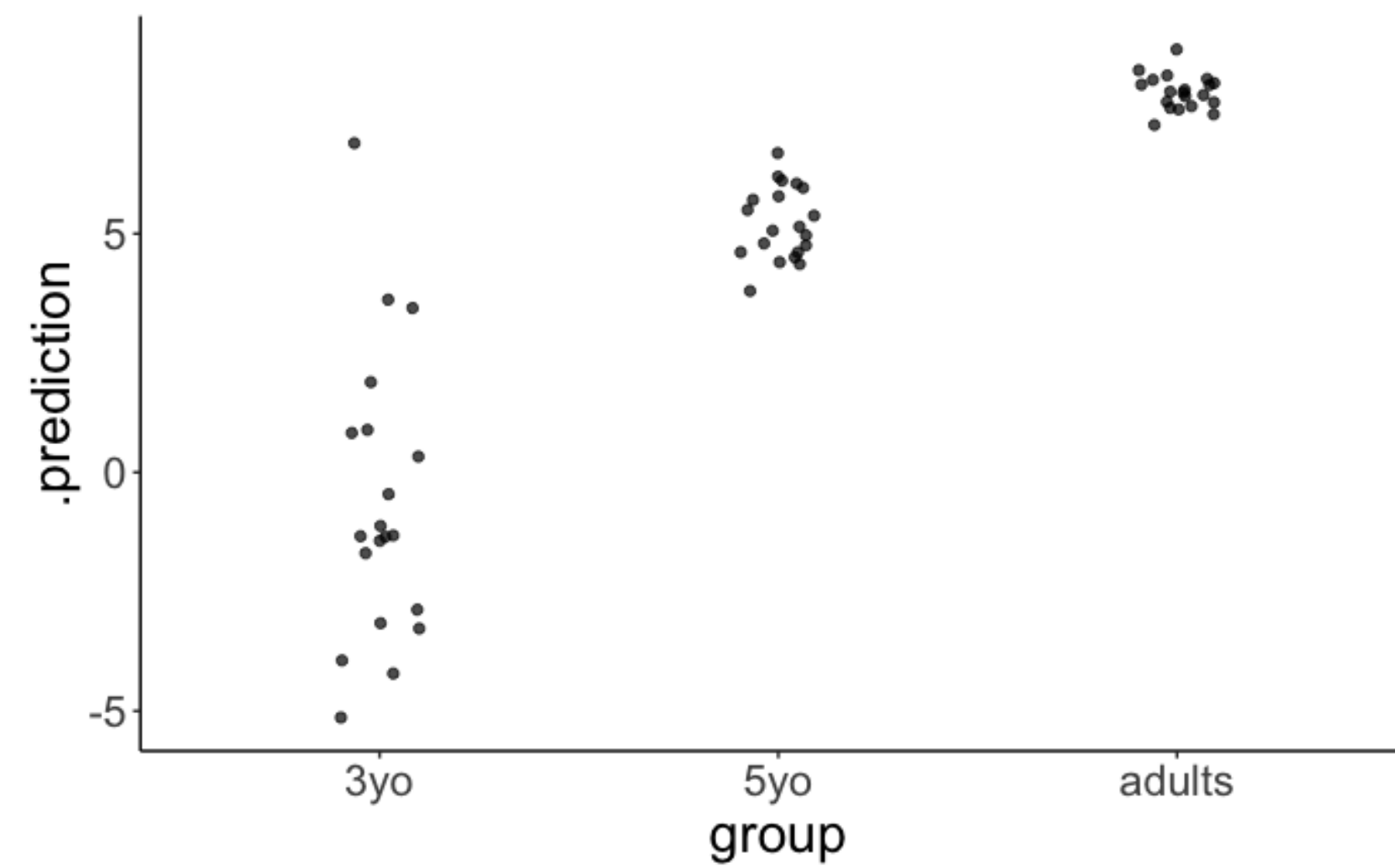
Unequal variance aka heteroscedasticity

```
1 df.variance %>%  
2   add_predicted_draws(model = fit.brm1,  
3                       n = 1) %>%  
4   ggplot(aes(x = group, y = .prediction)) +  
5   geom_jitter(height = 0,  
6               width = 0.1,  
7               alpha = 0.7)
```

original data



predicted data



these predictions look good!

A worked example

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<https://doi.org/10.1037/pspa0000445>

Inference From Social Evaluation

Zachary J. Davis¹, Kelsey R. Allen^{2, 3}, Max Kleiman-Weiner⁴, Julian Jara-Ettinger⁵, and Tobias Gerstenberg⁶

¹ Department of Psychology, Stanford University

² Department of Psychology, University of British Columbia

³ Department of Computer Science, University of British Columbia

⁴ Foster School of Business, University of Washington

⁵ Department of Psychology, Yale University

⁶ Department of Psychology, Stanford University

People have a remarkable ability to infer the hidden causes of things. From physical evidence, such as muddy footprints on the floor, we can figure out what happened and who did it. Here, we investigate another source of evidence: social evaluations. Social evaluations, such as praise or blame, are commonplace in everyday conversations. While such evaluations do not fully reveal what happened, they provide valuable clues. Across three experiments, we present situations where a person was praised or blamed, and participants' task is to use that information to figure out what happened. In Experiment 1, we find that people draw systematic inferences from social evaluations about situational factors, a person's actions, capabilities, and social roles. In Experiments 2 and 3, we develop computational models that generate praise and blame judgments by considering what causal role a person's action played, and what action they should have taken. Inverting these generative models of praise and blame via Bayesian inference yields accurate predictions about what inferences participants draw based on social evaluations.

Statement of Limitations

We looked at attributions of blame and praise, and what inferences people can draw from such attributions. As there is no clear normative standard for how such judgments and inferences should be made, we were not able to bonus participants based on performance. We assume that participants were sufficiently engaged in the task to produce systematic judgments. Participants acted as third-party judges in hypothetical scenarios, rather than producing behaviors that would be directly consequential to themselves or others. Across three experiments, we only considered a limited set of situations, and our experimental paradigms abstracted away many factors that make responsibility judgments challenging in the real world. We developed computational models of praise (in Experiment 2) and blame (in Experiment 3). While these models capture much of the variance in participants' judgments, it is possible that better models exist. Our models assume that people are making sophisticated inferences. However, it is possible that participants arrived at their judgments using heuristic shortcuts. All of our experiments relied on adult participant samples recruited online through Amazon Mechanical Turk and Prolific. This limits the potential generalizability of our findings.

Keywords: social evaluations, causality, inference, social cognition, blame

Andrew Todd served as action editor.

Tobias Gerstenberg <https://orcid.org/0000-0002-9162-0779>

All the data, study materials, and analysis code are available at https://github.com/cicl-stanford/inference_from_social_evaluation. Preregistrations for a subset of the experiments are available at <https://osf.io/yryjd6> (Experiment 1), <https://osf.io/s9dqw> (Experiment 2a), <https://osf.io/yz2nk> (Experiment 2b), and <https://osf.io/x37rj> (Experiment 3c).

Tobias Gerstenberg was supported by grants from Stanford's Human-Centered Artificial Intelligence Institute and from Cooperative AI. Julian Jara-Ettinger was supported by National Science Foundation Award BCS-2045778 awarded to Julian Jara-Ettinger. Some of the results from Experiment 3 have been reported in Allen et al. (2015; Experiment 3a and 3b) and in Davis et al. (2021; Experiment 3c). The authors thank the members of the Causality in Cognition Lab for feedback and discussion.

Zachary J. Davis played a supporting role in formal analysis,

investigation, methodology, software, visualization, and writing—review and editing and an equal role in data curation. Kelsey R. Allen played a supporting role in conceptualization, formal analysis, methodology, software, and writing—review and editing. Max Kleiman-Weiner played a supporting role in conceptualization, formal analysis, investigation, methodology, validation, visualization, and writing—review and editing. Julian Jara-Ettinger played a supporting role in conceptualization, software, and writing—review and editing. Tobias Gerstenberg played a lead role in conceptualization, formal analysis, funding acquisition, methodology, project administration, software, supervision, validation, visualization, writing—original draft, and writing—review and editing and an equal role in data curation.

Correspondence concerning this article should be addressed to Tobias Gerstenberg, Department of Psychology, Stanford University, 450 Jane Stanford Way, Building 420, Stanford, CA 94305, United States. Email: gerstenberg@stanford.edu

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2 Commits

code	.	last year
data	.	last year
docs	.	last year
figures	.	last year
.gitignore	.	last year
README.md	.	last year

README

Inference from social evaluation

This repository contains the experiments, data, analyses, and figures for the paper "[Inference from social evaluation](#)" by Zachary J. Davis, Kelsey R. Allen, Max Kleiman-Weiner, Julian Jara-Ettinger, and Tobias Gerstenberg (2025, *Journal of Personality and Social Psychology*).

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- [Preregistrations](#)
- [Repository structure](#)
- [CRedit author statement](#)

Introduction

People have a remarkable ability to infer the hidden causes of things. From physical evidence, such as muddy foot prints on the floor, we can figure out what happened and who did it. Here, we investigate another source of evidence: social evaluations. Social evaluations, such as praise or blame, are commonplace in everyday conversations. While such evaluations don't fully reveal what happened, they provide valuable clues. Across three experiments, we present situations where a person was praised or blamed, and participants' task is to use that information to figure out what happened. In Experiment 1, we find that people draw systematic inferences from social evaluations about situational factors, a person's actions, capabilities, and social roles. In Experiments 2 and 3 we develop computational models that generate praise and blame judgments by considering what causal role a person's action played, and what action they should have taken. Inverting these generative models of praise and blame via Bayesian inference yields accurate predictions about what inferences participants draw based on social evaluations.

A

Modeling framework

B

Example scenario

21

Feedback

How was the pace of today's class?

much
too
slow

a little
too
slow

just
right

a little
too
fast

much
too
fast

How happy were you with today's class overall?



What did you like about today's class? What could be improved next time?

Thanks